Sensitivity analysis of GRNmap and new features for GRNsight: open source software for dynamical systems modeling and visualization of small-scale gene regulatory networks in yeast

Poster GRNsight

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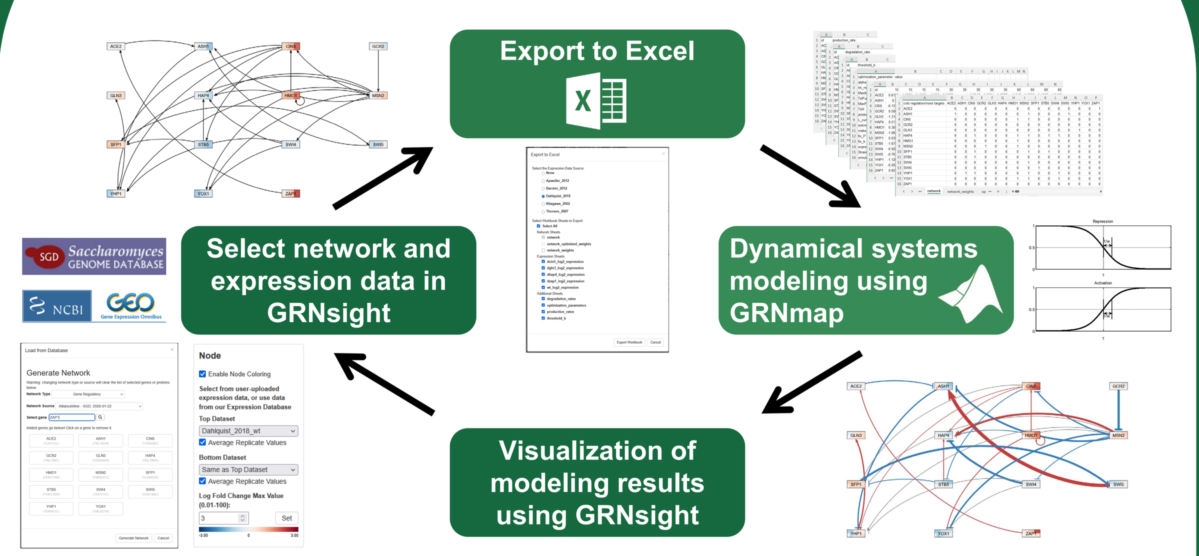
GRNmap

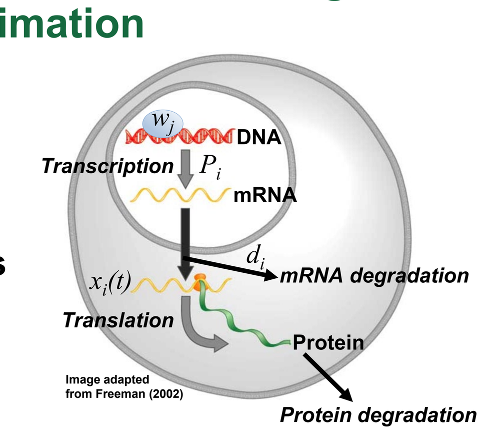
https://kdahlquist.github.io/GRNmap/ and https://dondi.github.io/GRNsight/

GRNmap and GRNsight are part of an integrated workflow to model and visualize GRNs in yeast



GRNmap: Gene Regulatory Network modeling and parameter estimation

$$\frac{dx_i(t)}{dt} = \frac{P_i}{1+\exp\left(-\left(\sum_j w_{ij}x_j(t)\right)-b_i\right)} - d_i x_i(t)$$



- **The change in mRNA expression of each gene in the network is modeled as *production – degradation* with a sigmoidal production function where:**
  - P_i is mRNA production rate for gene i.
  - d_i is the mRNA degradation rate for gene i.
  - w is the weight term, determining the magnitude of activation or repression of regulatory transcription factor(s) j on gene i.
  - b_i is the threshold of expression for each gene.

$$E = \alpha\|\theta\|^2 + \frac{1}{Q}\sum_{r=1}^{Q}\left[z^d(t_r) - z^c(t_r)\right]^2$$

- ***P*, *b*, and *w* parameters are estimated using a penalized least squares approach.**
  - α is determined empirically to find the best balance between minimizing both the penalty term (θ) and the error term (E).
  - When all three parameters are estimated, the total number of parameters is twice the number of genes plus the number of edges.
  - In the 15-gene, 28-edge network, there are 58 parameters being estimated, which are a lot given the amount of input data.
  - GRNmap can also:
    - hold P and b constant and only estimate w,
    - hold b constant and estimate P and w, or
    - hold P constant and estimate b and w.
- **The LSE:minLSE ratio is computed from the least squares error (E) from the estimation and the minimum theoretical error given the variance in the experimental expression data.**
  - The LSE:minLSE ratio can then be used as a metric to compare model performance for different networks.

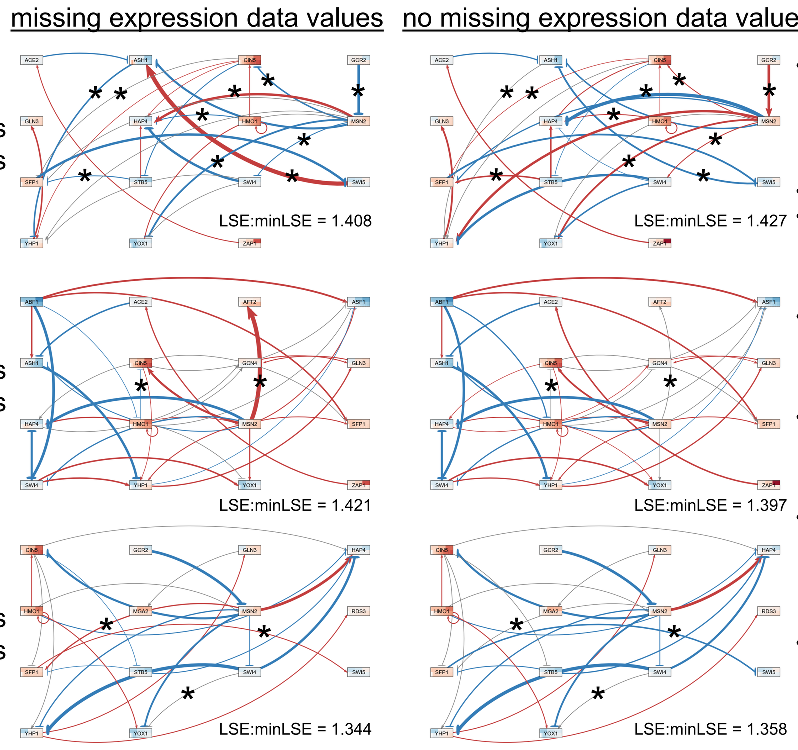
Three small candidate GRNs for regulating the cold shock response vary in sensitivity to small changes in input expression data

missing expression data values / no missing expression data values

15-genes 28-edges

16-genes 36-edges

14-genes 25-edges

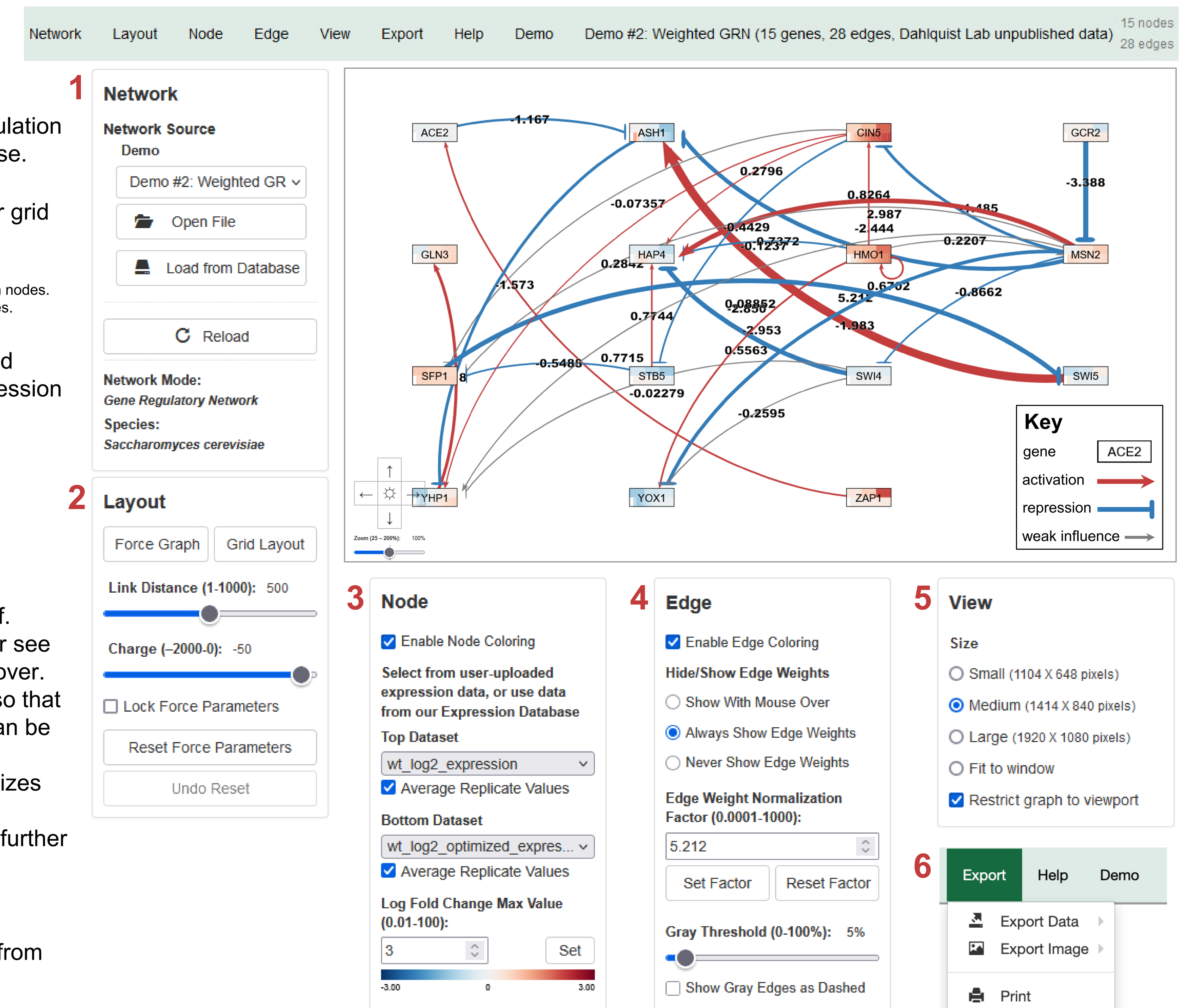


- **YEASTRACT-derived networks modeled with Dahlquist et al. (2018) expression data and visualized with GRNsight.**
- P, b, w all estimated.
- Log_2 expression data from wt yeast subjected to 15, 30, and 60 minutes of cold shock at 13°C.
- "missing": missing some expression values due to microarray data quality filters.
- "no missing": missing data values replaced with average value for that gene and timepoint.
- Top stripe in each node is actual log_2 fold change expression data for each timepoint, bottom stripe is model-simulated data.
- Edge thickness is normalized to a max weight of 5.212 (from top-left network).

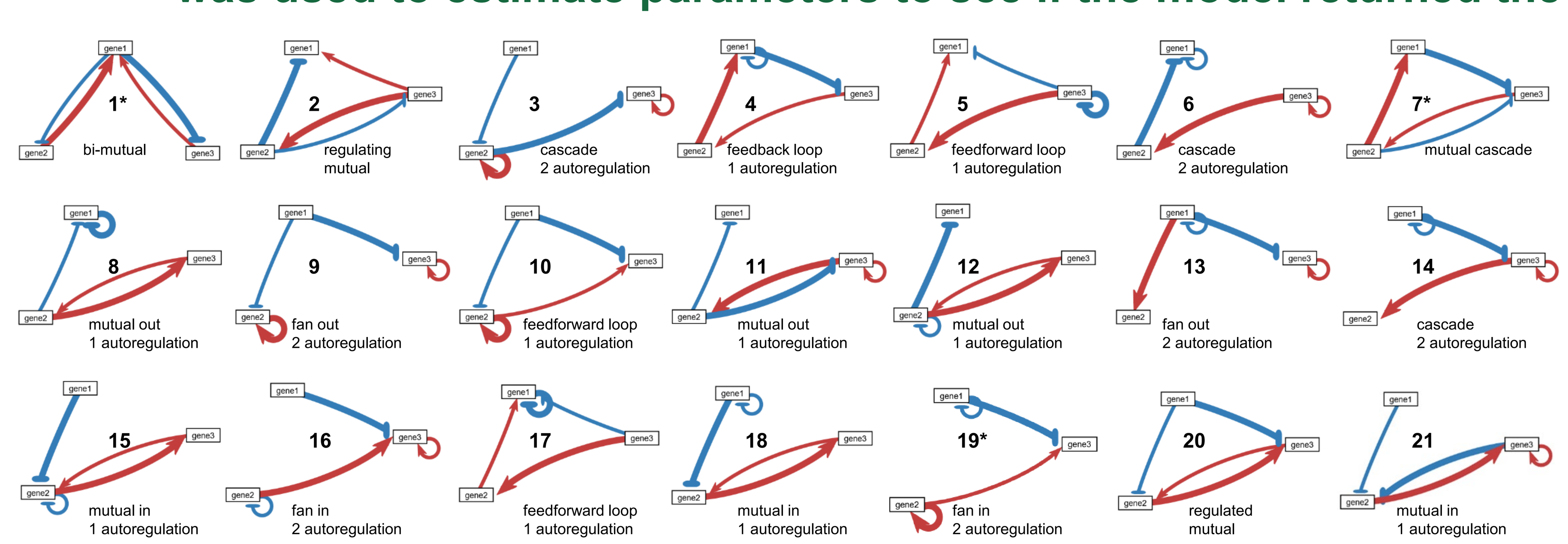
*Weight values that changed sign between estimations with missing input expression values and no missing values.
- The 15-gene, 28-edge network is far more sensitive to the input expression data than the bottom two networks.
- To investigate, we performed a parameter sensitivity analysis on smaller 3-gene, 4-edge synthetic networks.

GRNsight automatically lays out the network graph, coloring the edges based on weight values, and the nodes based on experimental or simulated expression data

1. **Network**
   - Demo files are provided.
   - Can import and export Excel, SIF, or GraphML files.
   - Can construct a network from SGD regulation or PPI data stored in a backend database.
2. **Layout**
   - Users can select a force graph layout or grid layout.
   - Force graph parameter sliders:
     - Link distance determines the minimum distance between nodes.
     - Nodes have a charge which repels or attracts other nodes.
3. **Node**
   - Coloring can be based on user-uploaded expression data or from published expression datasets stored in a backend database.
   - Datasets available:
     - Apweiler et al. 2012, GSE33098
     - Barreto et al. 2012, GSE24712
     - Dahlquist et al. 2018, GSE83656
     - Kitagawa et al. 2002, GSE9336
     - Thorsen et al. 2007, GSE6068
4. **Edge**
   - Edge coloring can be toggled on and off.
   - Buttons enable the user to always/never see edge weights or see them upon mouseover.
   - Allows user to set normalization factor so that edge thicknesses for different graphs can be rendered on the same scale.
   - Changing the gray threshold deemphasizes weaker edges in the visualization.
   - Presenting gray edges as dashed lines further deemphasizes weak edges.
5. **View**
   - Multiple viewport sizes available.
   - Graph bounding box can be separated from viewport.
   - Zooming and scrolling available.
6. **Export**
   - Network data export to SIF or GraphML or full data to GRNmap-compatible Excel workbook, with yeast mRNA degradation rates from Neymotin et al. (2014).
   - Image export to PNG, SVG, or PDF.



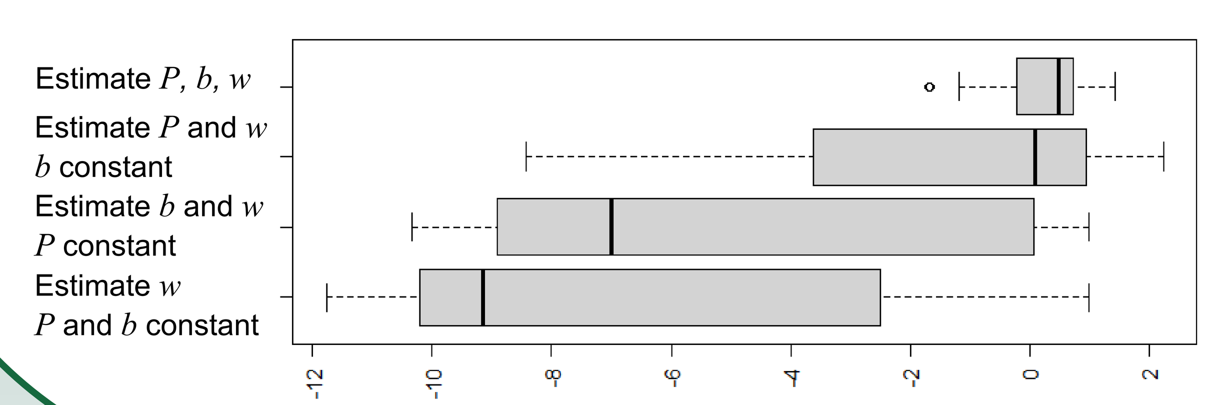
Model sensitivity was tested using all possible 3-gene, 4-edge "toy" networks. Simulated data was generated based on known parameters. Then the simulated data was used to estimate parameters to see if the model returned the known values.



- **Least squares error, LSE, was used to rank the performance of the networks relative to one another.**
- The error between the known weights and the estimated weights was computed as follows:

$$\mathrm{LSE} = \sum_{i=1}^{n}(w_i - w'_i)^2$$

where w_i are the actual weights and w'_i are the estimated weights.
- Ranking based on the sum of the LSEs under the four conditions.

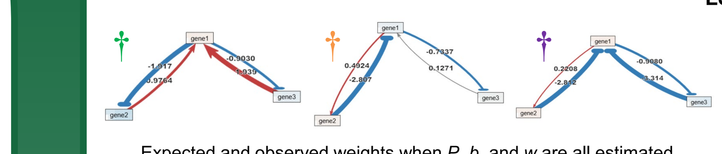


- When P and b were held constant, only the 4 weight parameters were estimated, compared to the 10 estimated when P, b, and w were all estimated.
- **The performance of the model when P and b were held constant and when only P was held constant were relatively similar, suggesting that the issue is estimating the production rate, not necessarily the number of parameters estimated.**

For the 3-gene, 4-edge "toy" networks, weight estimation precision is dependent on weight values

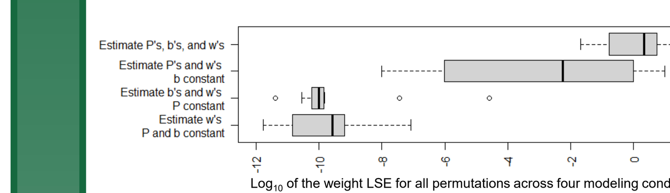
- Since weights were arbitrarily assigned (1, -1, 2, or -2), we also investigated if the rankings were solely based on network structure, or if they were also impacted by the weights assigned to the edges.
- All 24 permutations of the weight assignments to edges were modeled for the networks originally ranked 1st, 7th, and 19th to determine the impact of the weights.
- The LSE of the weights were calculated for each trial as before.

bi-mutual

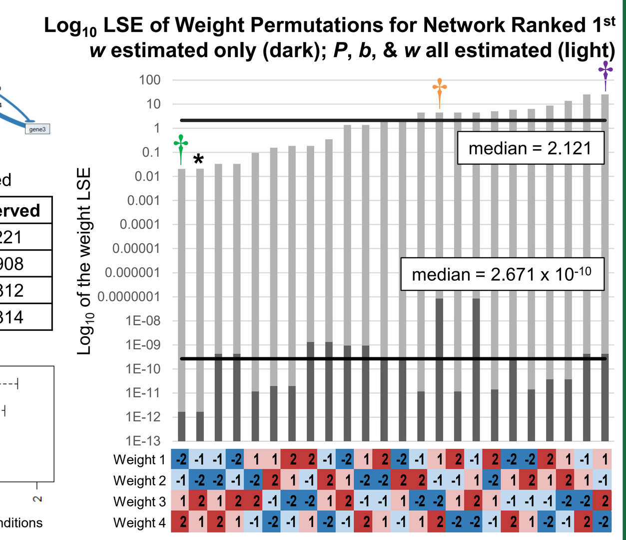


Expected and observed weights when *P*, *b*, and *w* are all estimated

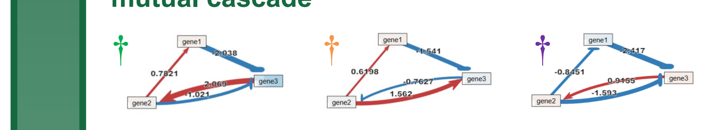
Expected	Observed	Expected	Observed	Expected	Observed
-2	-1.917	1	0.492	1	0.221
-1	-0.903	-1	-0.734	-1	-0.908
1	0.976	-2	-2.807	2	-2.812
2	1.939	2	0.127	-2	-3.314



Log_{10} LSE of Weight Permutations for Network Ranked 1st *w* estimated only (dark); *P*, *b*, & *w* all estimated (light)

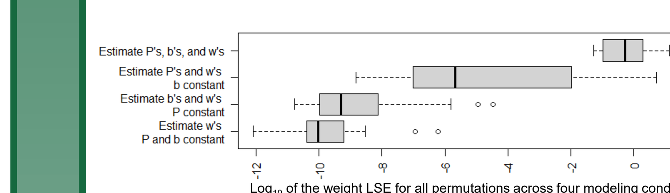


mutual cascade

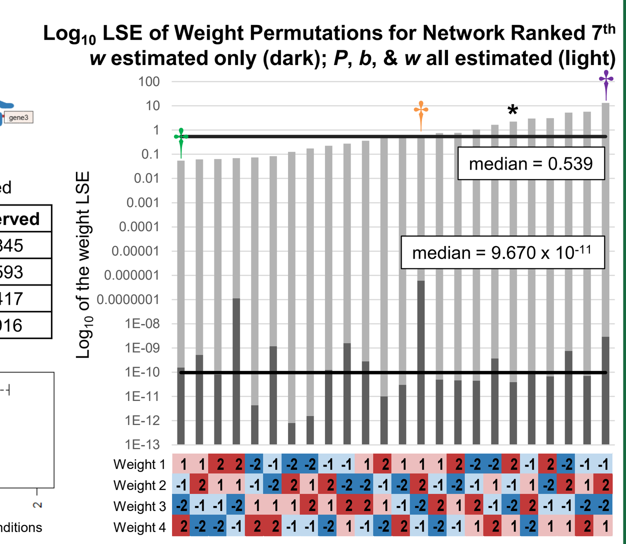


Expected and observed weights when *P*, *b*, and *w* are all estimated

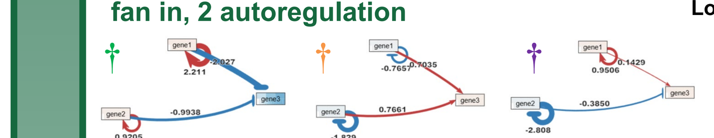
Expected	Observed	Expected	Observed	Expected	Observed
1	0.782	1	0.620	-1	-0.845
-1	-1.021	2	1.562	2	-1.593
-2	-2.038	-2	-1.541	-2	-2.417
2	2.069	-1	-0.763	1	0.916



Log_{10} LSE of Weight Permutations for Network Ranked 7th *w* estimated only (dark); *P*, *b*, & *w* all estimated (light)

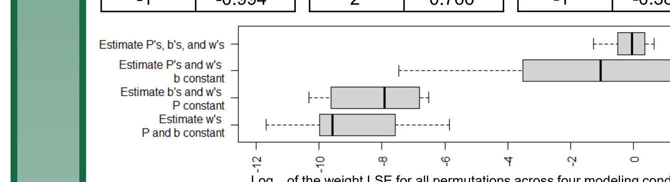


fan in, 2 autoregulation

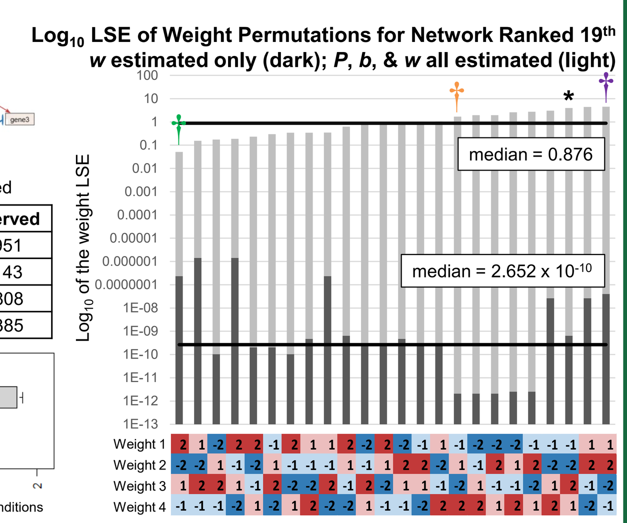


Expected and observed weights when *P*, *b*, and *w* are all estimated

Expected	Observed	Expected	Observed	Expected	Observed
2	2.211	-1	-0.766	1	0.951
-2	-2.027	1	0.704	2	0.143
1	0.921	-2	-1.829	-2	-2.808
-1	-0.994	2	0.766	-1	-0.385



Log_{10} LSE of Weight Permutations for Network Ranked 19th *w* estimated only (dark); *P*, *b*, & *w* all estimated (light)



- Generally, when only w is estimated, the LSE is very low ($10^{-12} – 10^{-6}$), so we ranked model performance when P, b, and w were all estimated, LSE = $10^{-2} – 10^{1}$ (see bar charts).
- The box and whisker plots show the same pattern that we saw previously where the median LSE is smaller in the two conditions where P is held constant.
- The mutual cascade family of networks performs better than the other two network families with smaller median LSEs for all estimation conditions.

Conclusions

- **GRNmap and GRNsight are part of an integrated workflow to model and visualize GRNs in yeast.**
- **Investigation of 3-gene, 4-edge "toy" networks showed that performance of the model estimation is influenced by a combination of network structure and edge weights.**

References

- Apweiler, E., Sameith, K., Margaritis, T., Brabers, N., van de Pasch, L., Bakker, L. V., ... & Kemmeren, P. (2012). Yeast glucose pathways converge on the transcriptional regulation of trehalose biosynthesis. *BMC genomics*, *13*(1), 1-14. https://doi.org/10.1186/1471-2164-13-239
- Barreto, L., Canadell, D., Valverde-Saubí, D., Casamayor, A., & Ariño, J. (2012). The short-term response of yeast to potassium starvation. *Environmental microbiology*, *14*(11), 3026-3042. https://doi.org/10.1111/j.1462-2920.2012.02887.x
- Dahlquist, K., Abdulla, H., Arnell, A., Baker, J., Carson, R., Citti, W., De Las Casas, S., Ellis, L., Entzminger, K., Entzminger, S., Fitzpatrick, B. G., Flores, S., Harmon, N., Hennessy, K., Herman, A., Hong, M., King, H., Kubeck, L., LaAnyane, O., … Zapata, B. (2018, January 22). NCBI. GEO Accession Display GSE83656. https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE83656
- Dahlquist, K.D., Fitzpatrick, B.G., Camacho, E.T., Entzminger, S.D., & Wanner, N.C. (2015) Parameter Estimation for Gene Regulatory Networks from Microarray Data: Cold Shock Response in Saccharomyces cerevisiae. *Bulletin of Mathematical Biology*, *77*(8), 1457-1492. https://doi.org/10.1007/s11538-015-0092-6
- Dahlquist, K.D., Dionisio, J.D.N., Fitzpatrick, B.G., Anguiano, N.A., Varshneya, A., Southwick, B.J., & Samdarshi, M. (2016) GRNsight: a web application and service for visualizing models of small- to medium-scale gene regulatory networks. *PeerJ Computer Science 2*:e85. https://doi.org/10.7717/peerj-cs.85
- Engel, S. R., Aleksander, S., Nash, R. S., Wong, E. D., Weng, S., Miyasato, S. R., ... & Cherry, J. M. (2025). Saccharomyces Genome Database: advances in genome annotation, expanded biochemical pathways, and other key enhancements. Genetics, *229*(3), iyae185. https://doi.org/10.1093/genetics/iyae185
- Freeman, S. (2002) Biological Science. Upper Saddle River, New Jersey: Prentice Hall.
- Kitagawa, E., Takahashi, J., Momose, Y., & Iwahashi, H. (2002). Effects of the pesticide thiuram: genome-wide screening of indicator genes by yeast DNA microarray. *Environmental science & technology*, *36*(18), 3908-3915. https://doi.org/10.1021/es015705v
- Teixeira, M. C., Viana, R., Palma, M., Oliveira, J., Galocha, M., Mota, M. N., ... & Monteiro, P. T. (2023). YEASTRACT+: a portal for the exploitation of global transcription regulation and metabolic model data in yeast biotechnology and pathogenesis. Nucleic acids research, *51*(D1), D785-D791. https://doi.org/10.1093/nar/gkac1041
- The Alliance of Genome Resources Consortium. (2024) Updates to the Alliance of Genome Resources central infrastructure. *Genetics*, *227*(1), iyae049. https://doi.org/10.1093/genetics/iyae049
- Thorsen, M., Lagniel, G., Kristiansson, E., Junot, C., Nerman, O., Labarre, J., & Tamás, M. J. (2007). Quantitative transcriptome, proteome, and sulfur metabolite profiling of the Saccharomyces cerevisiae response to arsenite. *Physiological genomics*, *30*(1), 35-43. https://doi.org/10.1152/physiolgenomics.00236.2006

Acknowledgments

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